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Fast Calibrated Additive Quantile Regression

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ABSTRACT

We propose a novel framework for fitting additive quantile regression models, which provides well-calibrated inference about the conditional quantiles and fast automatic estimation of the smoothing parameters, for model structures as diverse as those usable with distributional generalized additive models, while maintaining equivalent numerical efficiency and stability. The proposed methods are at once statistically rigorous and computationally efficient, because they are based on the general belief updating framework of Bissiri, Holmes, and Walker to loss based inference, but compute by adapting the stable fitting methods of Wood, Pya, and Säfken. We show how the pinball loss is statistically suboptimal relative to a novel smooth generalization, which also gives access to fast estimation methods. Further, we provide a novel calibration method for efficiently selecting the “learning rate” balancing the loss with the smoothing priors during inference, thereby obtaining reliable quantile uncertainty estimates. Our work was motivated by a probabilistic electricity load forecasting application, used here to demonstrate the proposed approach. The methods described here are implemented by the `ggam` R package, available on the Comprehensive R Archive Network (CRAN). Supplementary materials for this article are available online.

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Calibrated Bayes; Electricity load forecasting; Generalized additive models; Penalized regression splines; Quantile regression.

1. Introduction

Generalized additive models (GAMs, Hastie and Tibshirani 1990) are flexible and interpretable statistical models that are widely used in applied statistics, especially since the advent of efficient and stable methods for smoothing parameter selection and interval estimation in this model class (see, e.g., Wood 2000; Ruppert, Wand, and Carroll 2003; Fahrmeir, Kneib, and Lang 2004; Kim and Gu 2004; Wood 2017). The purpose of this work is to provide an equivalently useful framework for well-calibrated additive quantile regression models. Our methods are novel in that all smoothing parameters and other hyper parameters are estimated automatically using numerically robust and efficient methods which produce uncertainty estimates simultaneously with point estimates.

We were motivated by problems in electricity load forecasting. Electricité de France (EDF), France’s main electricity producer, has had considerable success using conventional GAMs for operational load forecasting. However, the whole conditional load distribution is rarely needed for production planning purposes, which focus mostly on tail estimates. This is because the loss function associated with forecasting errors is highly asymmetric, due to technical constraints (e.g., plant-specific start-up times or increasing fuel cost along the electricity production stack) and to the regulatory framework (e.g., monetary sanctions for over/under production). Further, the conditional distribution of the electricity load is typically highly skewed and time-dependent. At system-wide or substation level this problem is relatively mild, but new technologies (e.g., smart meters) are producing datasets where this issue is much more

extreme, due to the low level of aggregation. Full distributional modeling of the response distribution might be overly ambitious for these upcoming applications, hence it might be preferable to focus on estimating only the conditional quantiles most relevant to production planning or smart grid management.

To be usable in practical forecasting, additive quantile regression methods must have several properties: (1) the range of model structures available for modeling quantiles must be comparable to that available under conventional GAMs, otherwise the benefits of modeling quantiles may be offset by insufficient model flexibility; (2) smoothing and other tuning parameters must be selected automatically, otherwise the modeling process becomes too labor intensive and subjective for operational use; (3) uncertainty estimation has to be part of model estimation, since knowing forecast uncertainty is essential for operational use; and (4) methods must be sufficiently numerically efficient and robust for routine deployment. The work reported here started when two of the authors (YG and RN) were participating in the GEFCom2014 forecasting competition, and found that existing additive quantile regression method implementations failed to meet these requirements, forcing them to develop the ad hoc procedure described in Gaillard, Goude, and Nedellec (2016).

The framework developed in this article meets the four requirements by taking an empirical Bayesian approach to the general belief-updating framework of Bissiri, Holmes, and Walker (2016). Specifically, we represent smooth relationships between regressors and the quantile of interest using spline basis expansions, and impose Gaussian smoothing priors to

control model complexity. Random effects and parametric terms present no extra complication. By adopting a statistically improved smooth generalization of the usual quantile regression “pinball” loss (Koenker 2005), we are able to perform the computations required for belief updating of priors using the loss, and to estimate smoothing parameters, using the computational methods for general smooth modeling of Wood, Pya, and Säfken (2016). This allows us to achieve properties 1–4, provided that we can obtain the additional “learning rate” parameter required by the general belief updating framework. We show how to do this efficiently and automatically to achieve good calibration of the uncertainty estimates. Figure 1 provides some simple examples of the variety of models that our approach encompasses.

This is an advance relative to existing methods because, to our knowledge, pre-existing additive quantile regression methods fail to meet one or more of the four practically important requirements set above. For instance, the `quantreg` R package, which is based on the methods of Koenker (2013), requires users to select the smoothing parameters manually. The gradient boosting quantile regression method implemented by the `mboost` R package (Hothorn et al. 2010) requires users to manually choose the degrees of freedom used by each base model. In addition, `mboost` uses bootstrapping to estimate parameter uncertainty, while the approach proposed here quantifies uncertainty using computationally efficient asymptotic approximations. Yue and Rue (2011) and Waldmann et al. (2013) describe how to perform Bayesian inference for semiparametric additive quantile regression models. The first proposal is implemented in the `INLA` software (Martins et al. 2013), but the associated documentation discourages its use. In the second fitting is performed only via Markov chain Monte Carlo methods, which are much slower than the direct optimization methods proposed here. Further, it does not produce credible intervals with adequate frequentist properties for extreme quantiles, as mentioned below. The `vgam` R package (Yee 2008) provides a

method for fitting additive quantile regression models, but also in this case the complexity of the smooth terms is determined manually. The work of Lin et al. (2013) is not an alternative to what we propose here, because their focus is variable selection, rather than smoothing.

Quantile regression is traditionally based on the pinball loss (Koenker 2005), and not on a distributional model for the observations density, $p(y|x)$, which impedes direct application of Bayes’ rule. To circumvent this problem, Yu and Moyeed (2001) propose adopting an asymmetric Laplace (AL) model for $p(y|x)$, due to the equivalence between the AL negative log-density and the pinball loss. While Sriram, Ramamoorthi, and Ghosh (2013) proved that the resulting posterior concentrates around the true quantile, naively treating the AL density as an adequate probabilistic description of the data is problematic. In particular, Waldmann et al. (2013) showed that the resulting posterior credible intervals have poor frequentist calibration properties, especially for tail quantiles. Furthermore, this work will demonstrate that, in a nonparametric setting, selecting the scale parameter of the AL density using a likelihood based approach can lead to inaccurate quantile estimates (see Section 6.1). We solve both issues by adopting the beliefs updating framework of Bissiri, Holmes, and Walker (2016), and by coupling it with a calibration method which explicitly aims at achieving good frequentist properties.

This work also addresses the limitations implied by direct use of the pinball loss. The first issue is that this loss is piecewise linear, which impedes the use of computationally efficient fitting methods, designed to work with continuously differentiable, strongly convex functions. Yue and Rue (2011) and Oh, Lee, and Nychka (2012) addressed this problem by proposing smooth approximations to, respectively, the AL density and the pinball loss. The second issue (see Section 3) is that the pinball loss is statistically suboptimal relative to a smoothed generalization of the loss. Rather than smoothing the loss as little as possible (as in previous work) we therefore adopt the novel approach of using

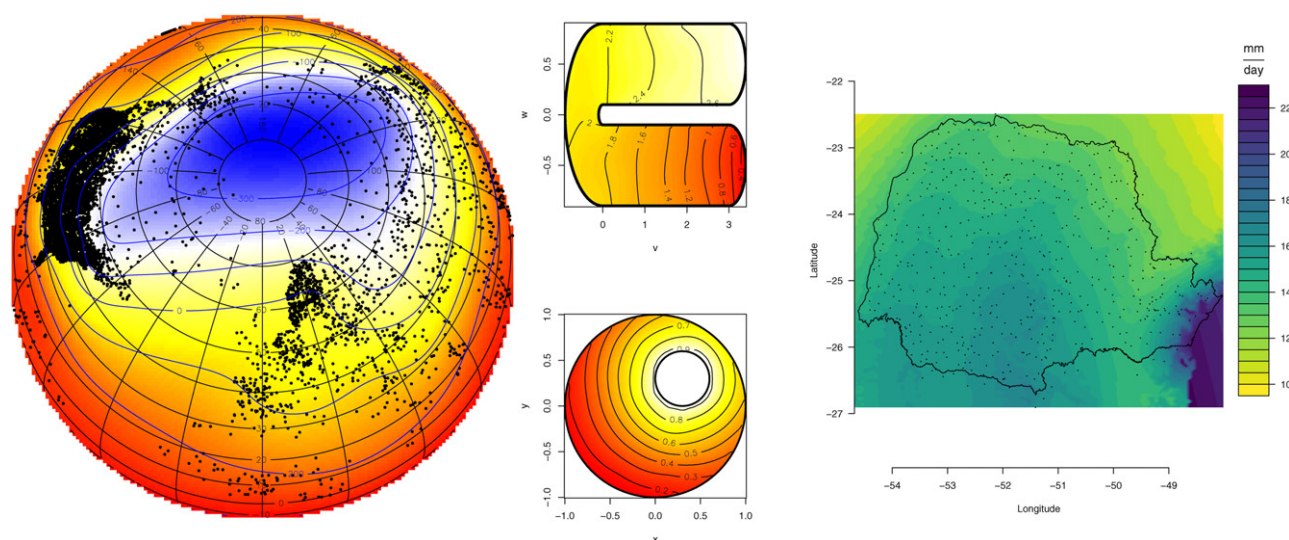


Figure 1. Examples of the smooth components that may be included in the additive quantile regression models fitted using the approach proposed here. Left: Effect of spatial location, defined using splines on the sphere, on quantile $\tau = 0.1$ of minimum daily temperatures, estimated using the Global Historical Climatology Network (GHCN) dataset (Menne et al. 2012). The Gulf Stream is visible. Center: Finite area spatial components, based on soap film smoothers, of two GAM fits for $\tau = 0.5$. The data are simulated. Right: Sum of the effects of spatial location, defined using an isotropic thin-plate spline basis, distance from the ocean and elevation, on quantile $\tau = 0.9$ of average weekly rainfall in Paraná state, Brazil. The dataset is available within the R-INLA R package (Lindgren and Rue 2015).

the degree of loss smoothness that minimizes the asymptotic MSE of the model regression coefficients.

The rest of the article is organized as follows. In Section 2, we briefly review additive quantile regression based on the pinball loss, while in Section 3, we describe the new loss function and show how its tuning parameter can be selected. In Section 4, we show how quantile regression can be set in a Bayesian context using the framework of Bissiri, Holmes, and Walker (2016), and we explain how additive quantile regression models can be fitted efficiently, if the new loss function is adopted. We propose a novel approach for posterior calibration in Section 5, and we test it on simulated examples in Section 6. In Section 7, we demonstrate the performance of the proposed approach for probabilistic electricity load forecasting.

2. Background on Additive Quantile Regression

Quantile regression aims at modeling the τ th quantile (where $\tau \in (0, 1)$) of the response, y , conditionally on a p -dimensional vector of covariates, $\mathbf{x}$. More precisely, if $F(y|\mathbf{x})$ is the conditional c.d.f. of y , then the τ th conditional quantile is $\mu = \inf\{y : F(y|\mathbf{x}) \geq \tau\}$. The τ th conditional quantile can also be defined as the minimizer of the expected loss

$$L(\mu|\mathbf{x}) = \mathbb{E}\{\rho_\tau(y - \mu)|\mathbf{x}\} = \int \rho_\tau(y - \mu)dF(y|\mathbf{x}), \quad (1)$$

w.r.t. $\mu = \mu(\mathbf{x})$, where

$$\rho_\tau(z) = (\tau - 1)z\mathbb{1}(z < 0) + \tau z\mathbb{1}(z \geq 0), \quad (2)$$

is the so-called pinball loss. Given a sample of size n , one approximates dF with its empirical version dF_n which, in a linear regression context, leads to the quantile estimator $\hat{\mu}(\mathbf{x}) = \mathbf{x}^\top \hat{\boldsymbol{\beta}}$ where

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n \rho_\tau\{y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\},$$

while $\mathbf{x}_i$ is the i th vector of covariates and $\boldsymbol{\beta}$ is the vector of regression coefficients.

In this work, we assume that $\mu(\mathbf{x})$ has an additive structure such as $\mu(\mathbf{x}) = \sum_{j=1}^m f_j(\mathbf{x})$, where the m additive terms can be fixed, random or smooth effects, defined in terms of spline bases. For instance, a marginal smooth effect could be $f_j(\mathbf{x}) = \sum_{k=1}^r \beta_{jk} b_{jk}(x_j)$, where β_{jk} are unknown coefficients and $b_{jk}(x_j)$ are known spline basis functions. Analogous expressions can be used to define joint or more complex smooths, such as those shown in Figure 1. The basis dimension r is typically chosen to be sufficiently generous that we can be confident of avoiding over-smoothing, but the actual complexity of f_j is controlled by a penalty on $\boldsymbol{\beta}_j$, designed to penalize departure from (some definition of) smoothness.

More specifically, let $\mu(\mathbf{x}_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$, where $\mathbf{x}_i$ is the i th row of the $n \times d$ design matrix $\mathbf{X}$, containing the spline basis functions evaluated at $\mathbf{x}_i$, and define the penalized pinball loss

$$V(\boldsymbol{\beta}, \boldsymbol{\gamma}, \sigma) = \sum_{i=1}^n \frac{1}{\sigma} \rho_\tau\{y_i - \mu(\mathbf{x}_i)\} + \frac{1}{2} \sum_{j=1}^m \gamma_j \boldsymbol{\beta}^\top \mathbf{S}_j \boldsymbol{\beta}, \quad (3)$$

where $\boldsymbol{\gamma} = \{\gamma_1, \dots, \gamma_m\}$ is a vector of positive smoothing parameters and $1/\sigma > 0$ is the so-called “learning rate,” which determines the relative weight of the loss and the penalty. As we will show later, it is possible to let σ depend on $\mathbf{x}$, which can be advantageous when the variance of y varies strongly with the covariates. The $\mathbf{S}_j$ ’s are positive semidefinite matrices and their purpose is penalizing the wiggleness of the corresponding effect f_j . In Section 4, we will show that, under the belief updating framework of Bissiri, Holmes, and Walker (2016), minimizing (3) w.r.t. $\boldsymbol{\beta}$ for fixed σ and $\boldsymbol{\gamma}$ leads to a maximum a posteriori (MAP) estimator, $\hat{\boldsymbol{\beta}}$. In particular, the $\mathbf{S}_j$ ’s can be seen as the precision matrices of a sequence of Gaussian priors on $\boldsymbol{\beta}$, centered at zero. The structure of each $\mathbf{S}_j$ is known and depends of the type of penalty used for f_j . The corresponding scaling factor, γ_j , is instead unknown and increasing it leads to a prior on f_j which is more concentrated around smoothness. Although the main challenge is the selection of $\boldsymbol{\gamma}$ and of σ , we first demonstrate that the pinball loss is statistically suboptimal, and can be improved in a manner that has the useful side effect of facilitating computation.

3. Optimally Smoothing the Pinball Loss

We consider the following generalization of the scaled pinball loss

$$\tilde{\rho}(y - \mu) = (\tau - 1) \frac{y - \mu}{\sigma} + \lambda \log(1 + e^{\frac{y - \mu}{\lambda \sigma}}), \quad (4)$$

where $\lambda > 0$ and the pinball loss is recovered as $\lambda \rightarrow 0$. Let $\text{Beta}(\cdot, \cdot)$ be the beta function. While the proposed loss can be seen as an instance of the function smoothing methods of Chen and Mangasarian (1995), we call (4) the Extended Log-F (ELF) loss because, upon normalization, $\exp(-\tilde{\rho})$ becomes

$$\tilde{\rho}_F(y - \mu) = \frac{e^{(1-\tau)\frac{y-\mu}{\sigma}} (1 + e^{\frac{y-\mu}{\lambda \sigma}})^{-\lambda}}{\lambda \sigma \text{Beta}\{\lambda(1-\tau), \lambda \tau\}}, \quad (5)$$

which is the p.d.f. of an extension of the log-F density of Jones (2008), as detailed in the Supplementary Material S1.1 (henceforth SM S1.1).

Let $\hat{\boldsymbol{\beta}}$ be the minimizer of the penalized ELF loss, the latter being defined analogously to (3). Define $h = \lambda \sigma$, $\mathbf{V} = \tau(1 - \tau)\mathbb{E}(\mathbf{x}_i \mathbf{x}_i^\top)$ and $u_i = y_i - \mu(\mathbf{x}_i)$, its p.d.f $f_{u|\mathbf{x}}(u|\mathbf{x})$ and c.d.f. $F_{u|\mathbf{x}}(u|\mathbf{x})$. Consider a simplified setting where the smoothing penalty is asymptotically dominated by the ELF loss as $n \rightarrow \infty$ and the true quantile $\mu^0(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}_0$, for some $\boldsymbol{\beta}_0$, which implies that the true quantile is linear in the design vector $\mathbf{x}$ of evaluated spline bases, so that $d = \dim(\mathbf{x})$ does not grow with n . In SM S1.2, we show that under the following further assumptions:

- (a) the pairs $\{\mathbf{x}_i, y_i\}$, with $i = 1, \dots, n$, are iid;
- (b) $\mathbb{P}(u_i < 0|\mathbf{x}_i) = \tau$ for almost all $\mathbf{x} \in \mathcal{X}$, the support of $\mathbf{x}$, and $i \in \{1, \dots, n\}$;
- (c) for almost all $\mathbf{x} \in \mathcal{X}$ and u in a neighborhood of zero, $f_{u|\mathbf{x}}(u|\mathbf{x})$ is three times continuously differentiable and there

exists a bounded function $C(\mathbf{x})$ such that $|f_{u|\mathbf{x}}^{(s)}(u|\mathbf{x})| \leq C(\mathbf{x})$ for $s \in \{1, 2, 3\}$, and $\mathbb{E}\{C(\mathbf{x})\|\mathbf{x}\|^2\} < \infty$;

(d) $h = O(n^{-\psi})$ with $1/5 < \psi < 2/5$;

(e) β_0 is the unique solution of $\mathbb{E}[\mathbf{x}_i\{\mathbb{1}(y_i - \mathbf{x}_i^\top \beta) > 0\} - 1 + \tau] = \mathbf{0}$ on $\beta \in \mathcal{B}$, the parameter space;

(f) $\Sigma_f = \mathbb{E}\{\mathbf{x}_i \mathbf{x}_i^\top f_{u|\mathbf{x}}(0|\mathbf{x}_i)\}$ and $\mathbb{E}(\mathbf{x}_i \mathbf{x}_i^\top)$ are nonsingular;

the following theorem holds:

Theorem 1. The asymptotic MSE of $\hat{\beta}$ is

$$\begin{aligned} \text{AMSE}(h) &= \mathbb{E}\{n(\hat{\beta} - \beta_0)(\hat{\beta} - \beta_0)^\top\} \\ &= \Sigma_f^{-1} \mathbf{V}^{\frac{1}{2}} \{\mathbf{I} + nh^4 \mathbf{B} \mathbf{B}^\top - h \mathbb{E}(\mathbf{A} \mathbf{A}^\top)\} \mathbf{V}^{\frac{1}{2}} \Sigma_f^{-1} \\ &\quad + O(h^2) + O(1/nh) + O(nh^6), \end{aligned} \quad (6)$$

where $\mathbf{B} = \pi^2 \mathbb{E}\{f_{u|\mathbf{x}}'(0|\mathbf{x}_i) \mathbf{V}^{-1/2} \mathbf{x}_i\}/6$ and $\mathbf{A} = f_{u|\mathbf{x}}(0|\mathbf{x}_i)^{1/2} \mathbf{V}^{-1/2} \mathbf{x}_i$.

Following Kaplan and Sun (2017), we seek the value of h that minimizes (6), after having discarded the higher order terms. Under assumption (d) this is justified, because the first term on the r.h.s. of (6) dominates the remainder. The AMSE of a linear combination $\sqrt{nc}^\top \hat{\beta}$ is $\mathbf{u}^\top \{\mathbf{I} + nh^4 \mathbf{B} \mathbf{B}^\top - h \mathbb{E}(\mathbf{A} \mathbf{A}^\top)\} \mathbf{u}$, where $\mathbf{u} = \mathbf{V}^{1/2} \Sigma_f^{-1} \mathbf{c}$, its minimizer being simply

$$h_c^* = \left\{ \frac{\mathbf{u}^\top \mathbb{E}(\mathbf{A} \mathbf{A}^\top) \mathbf{u}}{4n \mathbf{u}^\top \mathbf{B} \mathbf{B}^\top \mathbf{u}} \right\}^{\frac{1}{3}}.$$

If there is no interest in a specific $\mathbf{c}$, then it is preferable to minimize the average AMSE over several combinations. Following Kaplan and Sun (2017), we use d combinations $\mathbf{u}^1, \dots, \mathbf{u}^d$ such that $u_k^j = \mathbb{1}(k = j)$, which is equivalent to minimizing the trace of $nh^4 \mathbf{B} \mathbf{B}^\top - h \mathbb{E}(\mathbf{A} \mathbf{A}^\top)$, or the sum of its eigenvalues. As proved in SM S1.2, this leads to

$$h^* = \left\{ \frac{\mathbb{E}(\mathbf{A}^\top \mathbf{A})}{4n \mathbf{B}^\top \mathbf{B}} \right\}^{\frac{1}{3}}, \quad (7)$$

which, under the further assumption that the u_i 's are independent of the $\mathbf{x}_i$'s, becomes

$$\tilde{h}^* = \left\{ \frac{d}{n} \frac{9f_u(0)}{\pi^4 f_u''(0)^2} \right\}^{\frac{1}{3}}. \quad (8)$$

An alternative would be choosing $c_k^j = \mathbb{1}(k = j)$, for $j = 1, \dots, d$, but then the optimal bandwidth would not be invariant to the affine transformations of $\mathbf{x}$, as is the case for $\tilde{h}^*$.

The results in this section proved that the pinball loss ($\lambda = 0$) is not optimal, and that we should use the ELF loss with smoothness determined by $\tilde{h}^*$. This substitution of the smoothed loss greatly simplifies computation as it permits the use of smooth optimization methods for estimation. We now turn to the practical estimation of $\tilde{h}^*$.

3.1. Estimating $\tilde{h}^*$

Here we propose methods for estimation of $\tilde{h}^*$, considering also the treatment of heteroscedastic data. In particular, using a single bandwidth is inadequate in contexts where the variance of y strongly depends on $\mathbf{x}$. To address this issue, consider the location-scale model $y|\mathbf{x} \sim \alpha(\mathbf{x}) + \kappa(\mathbf{x})z$, where $\mathbb{E}(z|\mathbf{x}) = 0$ and $\text{var}(z|\mathbf{x}) = 1$. Assume that the z_i 's are iid, and let $\tilde{h}_z^*$ be the optimal bandwidth for regressing z on $\mathbf{x}$. Then the corresponding optimal bandwidth for y is $\tilde{h}^*(\mathbf{x}) = \tilde{h}_z^* \kappa(\mathbf{x})$, as can be verified using a simple change of variable argument. Given that in our context $\tilde{h}_z^* \kappa(\mathbf{x}) = \lambda \sigma$, one of the terms on the r.h.s. must depend on $\mathbf{x}$. We choose $\sigma(\mathbf{x}) = \sigma_0 \tilde{\sigma}(\mathbf{x})$, where σ_0 has been chosen using the methods of Section 5 and $n^{-1} \sum_{i=1}^n \tilde{\sigma}(\mathbf{x}_i) = 1$. This implies that $\lambda = n^{-1} \tilde{h}_z^* \sum_{i=1}^n \kappa(\mathbf{x}_i) / \sigma_0$. Under the above location-scale model, this leads to $1/\sigma(\mathbf{x}) \propto \omega(\mathbf{x}) = f_{u|\mathbf{x}}(0|\mathbf{x})$, which is the optimal weight function under the pinball loss $\sum_{i=1}^n \omega(\mathbf{x}_i) \rho_\tau\{y_i - \mu(\mathbf{x}_i)\}$ (Koenker 2005). Hence, while the baseline learning rate $1/\sigma_0$ is determined using the methods of Section 5, its $\mathbf{x}$ -dependent component $1/\tilde{\sigma}(\mathbf{x})$ is proportional to the optimal weighting function for quantile regression. It is reasonable to expect that the relative learning rate will be near-optimal also under the ELF loss, which is a close approximation to the pinball loss.

Our approach to loss bandwidth selection can be summarized in the following steps:

1. Estimate $\alpha(\mathbf{x})$ and $\kappa(\mathbf{x})$. Here we do this using a Gaussian GAM, where the mean and variance are estimated jointly using the methods of Wood, Pya, and Säfken (2016). We model $\alpha(\mathbf{x})$ using the same model used for $\mu_\tau(\mathbf{x})$, while for $\kappa(\mathbf{x})$ we typically use a simpler model (see Section 7 for an example).
2. Obtain the standardized residuals $z_i = \{y_i - \hat{\alpha}(\mathbf{x}_i)\} / \hat{\kappa}(\mathbf{x}_i)$, for $i = 1, \dots, n$, and get estimates $\hat{f}_z(\hat{\xi}_\tau)$ and $\hat{f}'_z(\hat{\xi}_\tau)$, where $\hat{\xi}_\tau$ is an estimate of the τ th quantile of z . We do this parametrically, by fitting the sinh-arsinh distribution of Jones and Pewsey (2009) to the z_i 's. If $|\hat{\xi}_\tau - \hat{\xi}_{\tau^*}| < \epsilon$, where $0 < \epsilon \ll 1$ and $\hat{\xi}_{\tau^*}$ is the mode of $\hat{f}_z$, we set $\hat{\xi}_\tau$ to $\hat{\xi}_{\tau+\delta}$ if $\hat{\xi}_\tau - \hat{\xi}_{\tau^*} > 0$ (or to $\hat{\xi}_{\tau-\delta}$ if $\hat{\xi}_\tau - \hat{\xi}_{\tau^*} < 0$), for some small $\delta > 0$. This is done to avoid dividing by $\hat{f}'_z(\hat{\xi}_\tau) \approx 0$ in (8).
3. Get the bandwidth $\tilde{h}^*(\mathbf{x}) = \tilde{h}_z^* \hat{\kappa}(\mathbf{x})$, where $\tilde{h}_z^*$ is obtained by plugging $\hat{f}_z$ and $\hat{f}'_z$ in (8), and by setting d to be the number of effective degrees of freedom used to model $\alpha(\mathbf{x})$ in step 1. Decompose the bandwidth into λ and $\sigma(\mathbf{x})$ as explained above.

Notice that the Gaussian GAM and the sinh-arcsinh density of steps 1 and 4 have to be fitted only once, so the resulting estimates can then be used to determine the bandwidths to be used for quantile regression at several probability levels. We used the sinh-arcsinh model, because it offers a good balance between flexibility and stability. The density used to model the distribution of the z_i 's does not matter asymptotically but, in finite samples, care is needed when estimating extreme quantiles, which we informally define as values of τ such that $\min\{n\tau, n(1 - \tau)\} < d$. In such cases the data are sparse and the estimated

quantile might depend strongly on the density model for the z_i 's, thus it is important to check for bias. A simple way of doing this is comparing the proportion of observations falling below the fitted quantile with its expected value, τ . Under such scenarios, one might also consider substituting the sinh-arcsinh density with a theoretically motivated tail model, such as the generalized Pareto distribution, but extreme quantile estimation is not the main focus of this article, hence we leave this idea for future work.

Adopting a location-scale model for loss smoothness selection avoids the issues of x -dependent bias and credible interval coverage in heteroscedastic scenarios (see Section 6.2 for an example of the latter problem). If further moments (e.g., the skewness) of the response distribution strongly depend on x , then better loss smoothness selection might be achieved by modeling those moments as well and plugging the resulting density estimates in (7). But, of course, this would increase the computational costs and modeling effort, hence we suggest adopting a more parsimonious location-scale model, which covers the main problems we have encountered in the context of electricity demand forecasting.

4. Model Fitting With Known Learning Rate

Having defined a smooth generalization of the pinball loss, and proposed methods for selecting its degree of smoothness, we describe a framework for fitting splines based additive quantile models. We first explain how to estimate the regression coefficients and the smoothing parameters, given σ_0 . Estimation of σ_0 is covered in Section 5.

4.1. Bayesian Quantile Regression via Coherent Belief-Updating

To set quantile regression in a Bayesian framework, we need to define a prior distribution on the regression coefficients, $p(\boldsymbol{\beta})$, and a mechanism for updating it to the corresponding posterior, $p(\boldsymbol{\beta}|\mathbf{y})$. Here we use the smoothing prior $\boldsymbol{\beta} \sim N(\mathbf{0}, \mathbf{S}^-)$, where $\mathbf{S}^-$ is an appropriate generalized matrix inverse of matrix $\mathbf{S}^y = \sum_{i=1}^m \gamma_i \mathbf{S}_j$, where $\mathbf{y}$ and the $\mathbf{S}_j$'s have been defined in Section 2. Given such a prior, direct application of Bayes' rule is impeded by the fact that we base quantile regression on the ELF loss, not on a probabilistic model for the observation density, $p(y|\boldsymbol{\beta})$, so the likelihood function is missing. Fortunately, this obstacle can be overcome by adopting the general belief-updating framework of Bissiri, Holmes, and Walker (2016), within which a prior distribution can be updated to produce a posterior while using a loss function, rather than a full likelihood, to connect model parameters to the data. Before applying it to quantile regression, we briefly outline the framework in its general form.

Assume that we are interested in finding the vector of model parameters $\boldsymbol{\beta}$ minimizing

$$\mathbb{E}\{L(\boldsymbol{\beta})\} = \int L(y, \boldsymbol{\beta}) f(y) dy, \quad (9)$$

where $L(\cdot, \cdot)$ is a general loss function and $f(y)$ is the p.d.f. of y . Suppose that we have a prior belief about $\boldsymbol{\beta}$, quantified by the prior density $p(\boldsymbol{\beta})$. Then Bissiri, Holmes, and Walker (2016)

argue that, given some data y , a coherent approach to updating $p(\boldsymbol{\beta})$ is the posterior

$$p(\boldsymbol{\beta}|y) = \frac{e^{-\frac{1}{\sigma} L(y, \boldsymbol{\beta})} p(\boldsymbol{\beta})}{\int e^{-\frac{1}{\sigma} L(y, \boldsymbol{\beta})} p(\boldsymbol{\beta}) d\boldsymbol{\beta}},$$

where $1/\sigma > 0$ is a "learning rate," determining the relative weight of the loss and the prior. When multiple samples, $\mathbf{y} = \{y_1, \dots, y_n\}$, are available this becomes

$$p(\boldsymbol{\beta}|\mathbf{y}) \propto e^{-\frac{1}{\sigma} \sum_{i=1}^n L(y_i, \boldsymbol{\beta})} p(\boldsymbol{\beta}), \quad (10)$$

where $\sum_{i=1}^n L(y_i, \boldsymbol{\beta})$ is an estimate of (9). Following Syring and Martin (2015) we call (10) the "Gibbs posterior" and its negative normalizing constant the "marginal loss."

Quantile regression, which we base on the ELF loss, fits squarely into this framework. In fact, the Gibbs posterior corresponding to such loss is

$$p(\boldsymbol{\beta}|\mathbf{y}) \propto \prod_{i=1}^n \tilde{p}_F\{y_i - \mu(\mathbf{x}_i)\} p(\boldsymbol{\beta}), \quad (11)$$

where $\tilde{p}_F$ is the ELF density (5), which implicitly depends on τ , λ , σ , and $\boldsymbol{\beta}$ (the latter via $\mu(\mathbf{x}_i)$). In the next section, we show how the regression coefficients can be estimated by maximizing the Gibbs posterior (11), given the smoothing parameters and learning rate.

4.2. Estimating the Regression Coefficients, $\boldsymbol{\beta}$, Given $\mathbf{y}$ and σ_0

Indicate with $\log\{\mu(\mathbf{x}_i), \sigma(\mathbf{x}_i)\}$ the i th element of the ELF loss (4) where, for fixed σ_0 , parameters λ and $\sigma(\mathbf{x}_i) = \sigma_0 \tilde{\sigma}(\mathbf{x}_i)$ have been selected using the methods of Section 3. Then, the negative Gibbs posterior log-density of $\boldsymbol{\beta}$ is proportional to the penalized loss

$$\tilde{V}(\boldsymbol{\beta}, \mathbf{y}, \sigma_0) = \sum_{i=1}^n \log\{\mu(\mathbf{x}_i), \sigma(\mathbf{x}_i)\} + \frac{1}{2} \sum_{j=1}^m \gamma_j \boldsymbol{\beta}^\top \mathbf{S}_j \boldsymbol{\beta}. \quad (12)$$

Hence, MAP estimates of the regression coefficients, $\hat{\boldsymbol{\beta}}$, can be obtained by minimizing (12), for fixed $\mathbf{y}$ and σ_0 . Given that the objective function is smooth and convex, this could be done efficiently using Newton algorithm, but a more stable solution can be obtained by exploiting orthogonal methods for solving least squares problems. In particular, notice that the minimizer of (12) corresponds to that of

$$\tilde{V}_D(\boldsymbol{\beta}, \mathbf{y}, \sigma_0) = \sum_{i=1}^n \text{Dev}_i\{\boldsymbol{\beta}, \sigma(\mathbf{x}_i)\} + \sum_{j=1}^m \gamma_j \boldsymbol{\beta}^\top \mathbf{S}_j \boldsymbol{\beta}, \quad (13)$$

where $\text{Dev}_i\{\boldsymbol{\beta}, \sigma(\mathbf{x}_i)\} = 2[\log\{\mu(\mathbf{x}_i), \sigma(\mathbf{x}_i)\} - \tilde{\mathbb{I}}]$ and $\tilde{\mathbb{I}}$ are, respectively, the i th component of the model deviance, based on (5), and the saturated loss, obtained by minimizing (4) w.r.t. μ . Then the regression coefficients can be estimated by penalized iteratively reweighted least squares (PIRLS), that is by iteratively minimizing

$$\sum_{i=1}^n w_i \{z_i - \mu_i\}^2 + \sum_{j=1}^m \gamma_j \boldsymbol{\beta}^\top \mathbf{S}_j \boldsymbol{\beta}, \quad (14)$$

where

$$z_i = \mu_i - \frac{1}{2w_i} \frac{\partial \text{Dev}_i}{\partial \mu_i}, \quad w_i = \frac{1}{2} \frac{\partial^2 \text{Dev}_i}{\partial \mu_i^2},$$

while $\mu_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ and $\text{Dev}_i = \text{Dev}_i\{\boldsymbol{\beta}, \sigma(\mathbf{x}_i)\}$.

4.3. Selecting the Smoothing Parameters, $\boldsymbol{\gamma}$, Given σ_0

A natural approach to selecting $\boldsymbol{\gamma}$, for fixed σ_0 , is minimizing the marginal loss

$$G(\boldsymbol{\gamma}, \sigma_0) = - \int \exp \left[- \sum_{i=1}^n \log\{\mu(\mathbf{x}_i), \sigma(\mathbf{x}_i)\} \right] p(\boldsymbol{\beta}|\boldsymbol{\gamma}) d\boldsymbol{\beta}, \quad (15)$$

which, as we noted in Section 4.1, is the negative of the normalizing constant of the Gibbs posterior. This is important from a computational point of view, because $G(\boldsymbol{\gamma}, \sigma_0)$ can be computed and minimized using efficient methods, originally developed to handle marginal likelihoods. In particular, $G(\boldsymbol{\gamma}, \sigma_0)$ involves an intractable integral which can be approximated using a Laplace approximation. This results in the Laplace approximate marginal loss (LAML) criterion

$$G_L(\boldsymbol{\gamma}, \sigma_0) = \frac{1}{2} \tilde{V}_D(\hat{\boldsymbol{\beta}}, \boldsymbol{\gamma}, \sigma_0) + n\tilde{l} + \frac{1}{2} \left[\log |\mathbf{X}^\top \mathbf{W} \mathbf{X} + \mathbf{S}^\gamma| - \log |\mathbf{S}^\gamma|_+ \right] - \frac{M_p}{2} \log(2\pi), \quad (16)$$

where $\hat{\boldsymbol{\beta}}$ is the minimizer of (12), $\tilde{l}$ is the saturated loss, $\mathbf{W}$ is a diagonal matrix such that $\mathbf{W}_{ii} = w_i$, M_p is the dimension of the null space of $\mathbf{S}^\gamma$ and $|\mathbf{S}^\gamma|_+$ is the product of its nonzero eigenvalues.

LAML can be efficiently minimized w.r.t. $\boldsymbol{\gamma}$, using an outer Newton algorithm. Numerically stable formulas for computing LAML and its derivatives are provided by Wood, Pya, and Säfken (2016). Importantly, the derivatives of $\hat{\boldsymbol{\beta}}$ w.r.t. $\boldsymbol{\gamma}$ are obtained by implicit differentiation which requires computing derivatives up to fourth order of the ELF loss w.r.t. μ . Notice that the w_i 's in (14) and (16) can be very close to zero when fitting quantile regression models based on the ELF density, hence obtaining reliable and numerically stable estimates requires modifying the PIRLS iteration and the computation of (16) and its derivatives. This more stable implementation is described in SM S2.

The Laplace approximation to the negative marginal log-likelihood based on the ELF density is obtained simply by adding $\sum_i \log[\lambda \sigma(\mathbf{x}_i) \text{Beta}\{\lambda(1-\tau), \lambda\tau\}]$ to (16), and it is possible to optimize it w.r.t. σ_0 as well as $\boldsymbol{\gamma}$. But σ_0 is confounded with the learning rate, so this cannot be justified by the Bissiri, Holmes, and Walker (2016) framework. Indeed, in Section 6, we refer to this approach as LAML selection of σ_0 and present examples of its failure in practice: it often produces inaccurate fits and poor interval calibration. Instead, Section 5 presents a calibration-based approach to the selection of σ_0 which, as the examples will show, alleviates both issues.

5. Calibrating σ_0

Here we propose a novel method for selecting σ_0 , which aims at obtaining approximately well-calibrated credible intervals for the quantile function, $\mu(\mathbf{x})$. In particular, let $C_\alpha\{\sigma_0, \mathbf{y}\}$ be the credible interval for $\mu(\mathbf{x})$, at level $\alpha \in (0, 1)$. The objective is selecting σ_0 so that

$$\mathbb{P}[\mu^0(\mathbf{x}) \in C_\alpha\{\sigma_0, \mathbf{y}\}] \approx \alpha, \quad (17)$$

for all α , where $\mathbb{P}$ is the objective probability measure, based on the data-generating process, and $\mu^0(\mathbf{x})$ is the true conditional quantile.

Let $\hat{\boldsymbol{\beta}}$ be the MAP estimate of the regression coefficients and define the covariance matrices $\mathbf{V} = (\mathcal{I} + \mathbf{S}^\gamma)^{-1}$ and $\tilde{\mathbf{V}} = (\mathcal{I} \tilde{\Sigma}_\gamma^{-1} \mathcal{I} + \mathbf{S}^\gamma)^{-1}$, where $\mathcal{I}$ is the Hessian of the unpenalized loss and $\tilde{\Sigma}_\gamma = \text{cov}[\nabla_{\boldsymbol{\beta}} \log\{\mu(\mathbf{x}), \sigma(\mathbf{x})\} |_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}}]$ w.r.t. $\mathbb{P}$. We select σ_0 by minimizing

$$\text{IKL}(\sigma_0) = n^{-1} \sum_{i=1}^n \left[\frac{\hat{v}(\mathbf{x}_i)}{v(\mathbf{x}_i)} + \log \frac{v(\mathbf{x}_i)}{\hat{v}(\mathbf{x}_i)} \right]^\zeta, \quad (18)$$

which is an estimate of the integrated Kullback–Leibler (IKL) divergence, that is

$$\begin{aligned} \text{IKL}(\sigma_0) &= \int \text{KL}[N\{\mu(\mathbf{x}), \tilde{v}(\mathbf{x})\}, N\{\mu(\mathbf{x}), v(\mathbf{x})\}]^\zeta p(\mathbf{x}) d\mathbf{x} \\ &\propto \int \left\{ \frac{\tilde{v}(\mathbf{x})}{v(\mathbf{x})} + \log \frac{v(\mathbf{x})}{\tilde{v}(\mathbf{x})} \right\}^\zeta p(\mathbf{x}) d\mathbf{x}, \end{aligned}$$

where $v(\mathbf{x}) = \mathbf{x}^\top \mathbf{V} \mathbf{x}$ and $\tilde{v}(\mathbf{x}) = \mathbf{x}^\top \tilde{\mathbf{V}} \mathbf{x}$ are the posterior variances of $\mu(\mathbf{x})$ under the two alternative covariance matrices for $\boldsymbol{\beta}$. ζ is a positive constant and $N(\cdot, \cdot)$ indicates the normal distribution. In (18) $\tilde{\mathbf{V}}$ is replaced by $\hat{\tilde{\mathbf{V}}} = (\mathcal{I} \hat{\tilde{\Sigma}}_\gamma^{-1} \mathcal{I} + \mathbf{S}^\gamma)^{-1}$, where $\hat{\tilde{\Sigma}}_\gamma$ is the regularized estimator proposed in SM S3.2. Objective (18) is deterministic and one dimensional, hence it can be efficiently minimized using standard root-finding methods, such as bisection. In our experience, the objective is generally smooth and it has a unique minimum. Decreasing (increasing) σ_0 leads to wigglier (smoother) fits and increases (decreases) $\tilde{v}(\mathbf{x})/v(\mathbf{x})$.

Our approach is motivated as follows. Notice that the Gibbs posterior can be seen as a posterior based on the misspecified parametric likelihood, formed by the ELF density. Müller (2013) proves that, while the posterior of misspecified models is asymptotically Gaussian with mean vector $\hat{\boldsymbol{\beta}}$ and covariance matrix $\mathbf{V}$, this posterior is asymptotically worse, in terms of frequentist risk, than a posterior having “sandwich” covariance $\tilde{\mathbf{V}}$. Given that credible intervals can be derived within a decision-theoretic framework by adopting an appropriate loss function (see, e.g., Robert 2007, sec. 5.5.3), and that Müller’s work considers general losses, it is clear that the intervals based on $\tilde{\mathbf{V}}$ should have better asymptotic frequentist properties. Hence, we minimize (18) w.r.t. σ_0 so that the marginal posterior distribution of $\mu(\mathbf{x})$, which is based on $\mathbf{V}$, is as close as possible to that based on $\tilde{\mathbf{V}}$. We choose $\zeta = 1/2$, and in general we suggest setting $0 < \zeta < 1$, to make IKL more robust to the occasional large discrepancies between $\tilde{v}(\mathbf{x})$ and $v(\mathbf{x})$, which can occur where the design points are sparse. Notice that the KL divergence is asymmetric, hence IKL is not invariant to the ordering of $v(\mathbf{x})$

and $\tilde{v}(\mathbf{x})$. We prefer the ordering used here, because it penalizes under-coverage ($\tilde{v}(\mathbf{x}) > v(\mathbf{x})$) more than over-coverage ($\tilde{v}(\mathbf{x}) < v(\mathbf{x})$).

Recall that we are minimizing the discrepancy between the marginal posteriors for $\mu(\mathbf{x})$ based on $\mathbf{V}$ and $\tilde{\mathbf{V}}$ because the latter offers better asymptotic frequentist properties. However, Müller (2013) clarifies that adopting $\tilde{\mathbf{V}}$ does not lead to a posterior achieving the lowest possible asymptotic risk. Hence, it is reasonable to expect that intervals based on the true marginal variance of $\hat{\mu}(\mathbf{x}) = \mathbf{x}^\top \hat{\boldsymbol{\beta}}$ under $\mathbb{P}$ would offer better coverage, especially in small samples. To provide such an alternative to the sandwich estimator, SM S3.1 proposes a bootstrapping procedure for estimating a different IKL loss, where an estimate of $\text{var}\{\hat{\mu}(\mathbf{x})\}$ under $\mathbb{P}$ substitutes $\tilde{v}(\mathbf{x})$.

Motivated by the nonparametric spline-based context considered here, we have chosen to explicitly calibrate the posterior of $\mu(\mathbf{x})$, rather than that of $\boldsymbol{\beta}$. However, the following argument suggests that the calibration procedure proposed here should lead to approximately calibrated intervals for $\boldsymbol{\beta}$ too. Assume that $\mathbf{X}$ is a $n \times d$ full rank matrix and suppose that minimizing the IKL loss leads to a value of σ_0 such that $\mathbf{x}_i^\top \mathbf{V} \mathbf{x}_i = \mathbf{x}_i^\top \tilde{\mathbf{V}} \mathbf{x}_i$, for $i = 1, \dots, n$. Then the properties of Kronecker products lead to $\mathbf{X} \otimes_r \mathbf{X} \text{vec}(\mathbf{V}) = \mathbf{X} \otimes_r \mathbf{X} \text{vec}(\tilde{\mathbf{V}})$, where $\otimes_r$ indicates the row-wise Kronecker product, such that the i th row of $\mathbf{X} \otimes_r \mathbf{X}$ is $\mathbf{x}_i \otimes \mathbf{x}_i$. Given that $\mathbf{X}$ is of full rank d , then the symmetry of $\mathbf{V}$ and the fact that $\text{rank}(\mathbf{X} \otimes_r \mathbf{X}) \geq d(d+1)/2$ imply that $\mathbf{V} = \tilde{\mathbf{V}}$.

Müller (2013) proved that $\tilde{\mathbf{V}}$ achieves a lower asymptotic frequentist risk than $\mathbf{V}$ in a setting where the prior is increasingly dominated by the likelihood as n increases, so that the asymptotic variance of $\hat{\boldsymbol{\beta}}$ does not depend on the prior. In a penalized cubic regression spline context such dominance occurs when the spline basis dimension $d = O(n^\alpha)$ for $\alpha < 1/5$. This includes the regime considered by Kauermann, Krivobokova, and Fahrmeir (2009), when demonstrating the statistical validity of GAM inference based on Laplace approximate marginal likelihood smoothing parameter estimation. However, other regimes are also possible (e.g., Claeskens, Krivobokova, and Opsomer 2009), and the question of relative risk is then open.

6. Simulated Examples

Before applying the proposed quantile regression framework to load forecasting, we test it on two simulated examples. In particular, in Section 6.1 we fit an additive quantile model to

homoscedastic data with $\sigma(\mathbf{x}) = \sigma_0$, while in Section 6.2 we consider heteroscedastic data, where adequate interval coverage can be achieved only by letting the learning rate and the ELF loss bandwidth vary with $\mathbf{x}$.

6.1. An Additive Example

Consider the following additive model

$$y_i = x_i + x_i^2 - z_i + 2\sin(z_i) + 0.1v_i^3 + 3\cos(v_i) + e_i, \quad (19)$$

where $e_i \sim \text{gamma}(3, 1)$, $x_i \sim \text{unif}(-4, 4)$, $z_i \sim \text{unif}(-8, 8)$, and $v_i \sim \text{unif}(-4, 4)$. We aim at estimating the conditional quantiles corresponding to $\tau = 0.01, 0.05, 0.5, 0.95$, and 0.99 . Hence, we fit an additive quantile regression model for each τ , using the ELF loss. We determine the loss bandwidth as in Section 3, where the $\mathbb{E}(y|\mathbf{x}) = \alpha(\mathbf{x})$ and $\text{var}(y) = \kappa^2$ are estimated using a Gaussian GAM. Fitting this model has a negligible impact on the computational cost, as it has to be done only once, before calibrating σ_0 . We select σ_0 either by minimizing LAML w.r.t. both σ_0 and $\boldsymbol{\gamma}$, or by the calibration method of Section 5. We consider two versions of the latter, one based on the sandwich covariance matrix $\tilde{\mathbf{V}}$, the other on the bootstrapping routine of SM S3.1. We also include quantile regression by gradient boosting, as implemented in the `mbboost` R package (Hothorn et al. 2010).

We simulate 100 datasets from (19), using either $n = 10^3$ or $n = 10^4$, and we fit an additive model for each τ using each approach. The fitted model includes a smooth effect for each covariate, based on cubic regression splines bases of rank 30. The boosting approach requires also selecting the degrees of freedom of each effect, which we set to 6. The number of boosting iterations was selected by minimizing the out-of-bag empirical risk, based on the pinball loss and on 100 bootstrap datasets. The boosting step size was equal to 0.1 when $n = 10^3$ and 1 when $n = 10^4$. To select σ_0 by posterior calibration, we minimized the estimated IKL loss using Brent's method (Brent 2013). The bootstrap version of the procedure was based on 100 bootstrap samples.

Table 1 reports the average RMSE ($[n^{-1} \sum_i \{\hat{\mu}(\mathbf{x}_i) - \mu^0(\mathbf{x}_i)\}^2]^{1/2}$). When $n = 10^3$, the RMSEs achieved by the two calibration approaches are strictly lower than those achieved by boosting. The performance of boosting is closer to that of our method when $n = 10^4$, which suggests that, for fixed model complexity, the advantage of using a smooth loss is inversely proportional to the amount of data available. LAML selection of σ_0 leads to worse results relative to our method, especially

Table 1. Additive example: mean(SD) of the RMSEs between true and estimated quantiles, for each quantile and method for $n = 10^3$ (top rows) and $n = 10^4$ (bottom rows).

τ	0.01	0.05	0.5	0.95	0.99
CAL Boot	0.273 (0.04)	0.237 (0.03)	0.309(0.04)	0.722(0.1)	1.104(0.23)
CAL Sand	0.274(0.04)	0.237 (0.03)	0.303 (0.04)	0.717 (0.1)	1.097 (0.22)
LAML	0.284(0.03)	0.249(0.03)	0.307(0.04)	0.926(0.14)	1.284(0.16)
BOOST	0.369(0.08)	0.272(0.04)	0.321(0.05)	0.814(0.11)	1.674(0.38)
CAL Boot	0.102(0.01)	0.093(0.01)	0.125(0.01)	0.314(0.03)	0.543(0.08)
CAL Sand	0.1 (0.01)	0.092 (0.01)	0.123 (0.01)	0.307 (0.03)	0.535 (0.08)
LAML	0.113(0.01)	0.1(0.01)	0.126(0.01)	0.406(0.04)	0.897(0.09)
BOOST	0.107(0.01)	0.094(0.01)	0.123 (0.01)	0.307 (0.04)	0.561(0.08)

NOTE: The lowest RMSE(s), for each sample size and quantile, is bold.

for the highest quantiles. Hence, in this example the calibration procedure based on $\tilde{V}$ leads to quantile estimates that are as accurate as those produced by bootstrap based calibration, and more accurate than those obtained by boosting, and much cheaper to compute. On an Intel 2.50 GHz CPU, calibrating σ_0 using $\tilde{V}$ takes around 1.4 sec for $n = 10^3$ and 11 sec for $n = 10^4$, when $\tau = 0.5$. Under bootstrapping, the calibration takes 13 sec and 126 sec, while selecting the number of boosting steps takes around 134 sec (2000 steps) and 150 sec (550 steps). However, selecting the number of boosting step takes much longer for $\tau = 0.01$: 0.6 hr (3×10^4 steps) and 0.5 hr (6000 steps). For the same quantile $\tilde{V}$ -based calibration takes 3.5 sec and 20 sec, while the bootstrap version takes 45 sec and 350 sec. In practice `mboost`'s computing times are longer, as the cross-validation needs to run beyond the optimal step size, which is not known in advance. Notice that the leading order computational cost of the methods developed here is $O(nd^2)$ with $O(nd)$ storage requirements. Under $\tilde{V}$ -based calibration and with $n = 10^5$, estimating the same five quantiles of model (19) takes around 15 min on the hardware mentioned above.

Figure 2 shows the empirical coverage, at 95, 75, and 50% level, achieved by the credible intervals for $\mu(x)$, using calibration or LAML to select σ_0 . We do not check the coverage achieved by gradient boosting, because analytic formulas are unavailable and confidence intervals must be obtained by bootstrapping with each bootstrap replicate as computationally expensive as the original fit. Notice that the coverage achieved using LAML for selecting σ_0 is well below nominal levels for most quantiles, and does not improve as n increases. Instead, under bootstrap or sandwich-based calibration, coverage improves with n . In particular, the bootstrap version attains close to nominal coverage for most quantiles when $n = 10^4$, and almost perfect coverage for $n = 5 \times 10^4$. The sandwich version shows a similar improvement but, for large n , it selects larger values of σ_0 than the bootstrap version, which leads to slightly wider intervals. The bootstrap-based calibration selects a lower value of σ_0 because it minimizes a version of the IKL criterion which takes into account the bias of the fit (see SM S3.1 for details), which is directly proportional to σ_0 .

6.2. An Heteroscedastic Example

Here we consider the following heteroscedastic data generating process

$$y_i \sim \text{SkewNorm}\{\xi(x_i), \omega(x_i), \theta\}, \quad \xi(x_i) = x_i + x_i^2, \\ \omega(x_i) = 1.5 + \sin(2x_i), \quad \theta = 4, \quad (20)$$

where ξ , ω , and θ are the location, scale, and shape parameters of the skew-normal distribution (Azzalini 1985), while $x_i \sim \text{unif}(-4, 4)$. We simulate $n = 2000$ data points from (20) and we fit quantile models for the median and the 95th percentile. In particular, we consider a simplified model where $\sigma(x) = \sigma_0$, and a full model where the learning rate and the loss bandwidth vary with x . As explained in Section 3, this requires fitting a location-scale model to estimate the conditional mean and variance of y . We use a Gaussian GAM where both the mean and the variance of y depend on x , which we fit using the methods of Wood, Pya, and Säfken (2016). We model the quantiles using cubic regression spline bases of rank 30, and we adopt the same basis for the mean and variance of the Gaussian GAM.

The first two columns in Figure 3 compare nominal and empirical coverage of credible intervals for $\mu(x)$, obtained by fitting 5000 datasets simulated from (20) using $\tilde{V}$ -based calibration. Clearly, the simplified model provides unreliable intervals even at the median, while the intervals of the full model are much closer to nominal levels. Figure 3 also shows the fits for quantiles $\tau = 0.1, 0.2, \dots, 0.9$ of the motorcycle dataset (Silverman 1985). These were obtained using an adaptive P-spline basis of rank 20 to model the quantile curves and the mean of the Gaussian GAM, and a thin-plate spline basis of rank 10 for the conditional variance.

7. Probabilistic Load Forecasting

GAMs have proved highly successful at EDF, because they can capture the complex relations existing between electricity load and several meteorological, economic and social factors, while retaining a high degree of interpretability, which is critically important during exceptional events, when manual intervention might be required. However, the cost structure relevant to an electrical utility implies that only certain conditional quantile

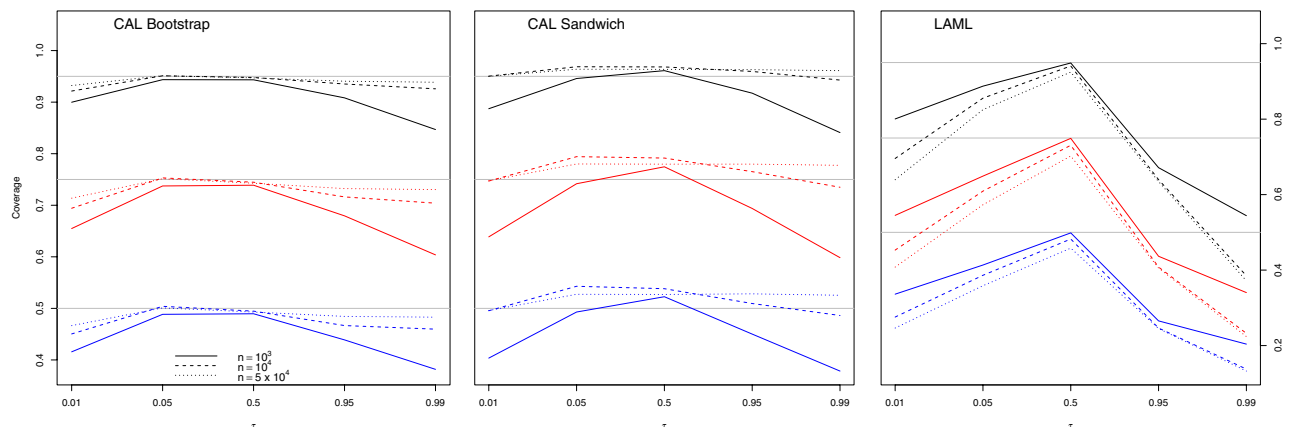


Figure 2. Additive example: Empirical coverage achieved by selecting σ_0 by calibration (using either the sandwich covariance $\tilde{V}$ or bootstrapping) or by LAML minimization, for each τ and sample size n . The horizontal lines indicate the nominal coverage levels.

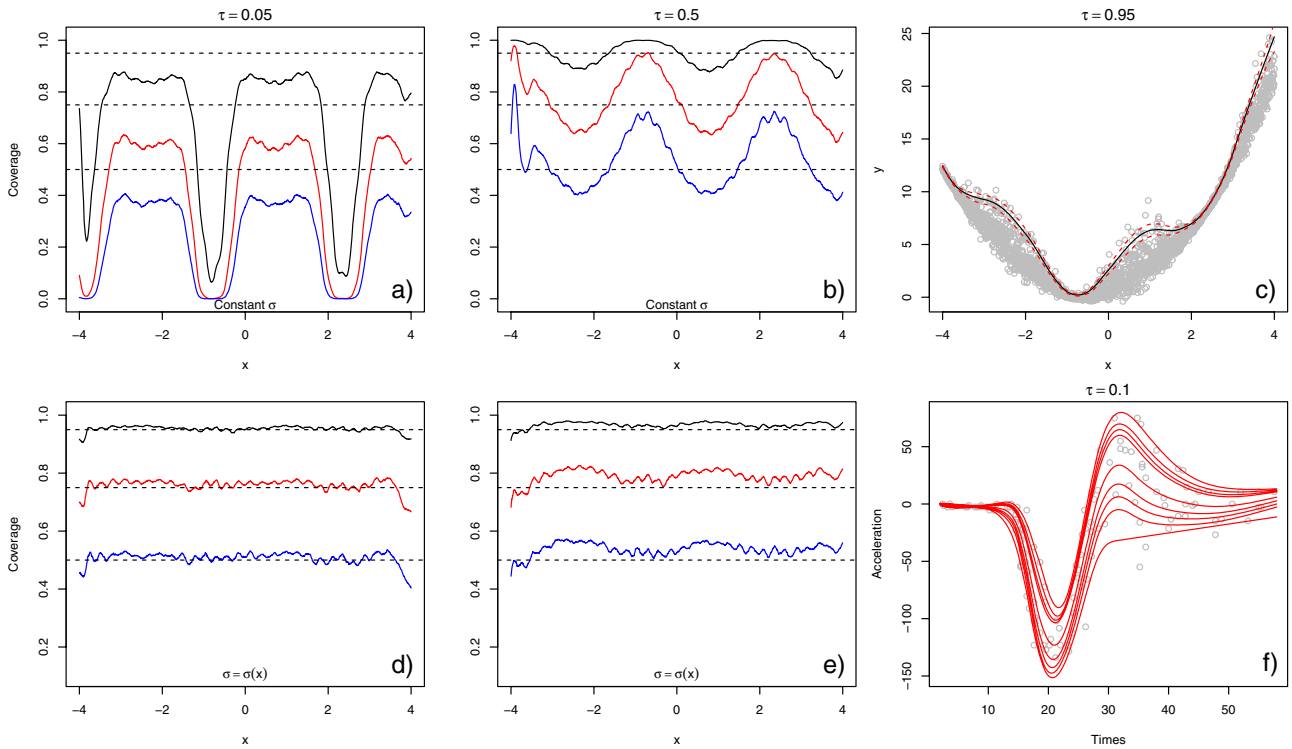


Figure 3. (a, b, d, and e) Nominal (dashed) versus empirical (solid) coverage at 50, 75, and 95% level, using a simplified (a and b) or full quantile model (d and e). (c) Full fit using $\tau = 0.95$ with data from model (20). (f) Full fits for several quantiles of the motorcycle dataset.

estimates are of high operational interest. This, and the difficulty of finding a parametric model for the load distribution that holds at several levels of aggregation, makes of semiparametric quantile regression an attractive alternative to traditional GAMs.

In this section we consider the three datasets shown in Figure 4. The first is the dataset used in the load forecasting track of the Global Energy Competition 2014 (GEFCom2014). This covers the period between January 2005 and December 2011, and it includes half-hourly load consumption and temperatures. The other two datasets contain half-hourly electricity demand from the UK and French grids. The first covers the period between January 2011 and June 2016, the second between January 2013 and December 2017. We integrate them with hourly temperature data from the National Centers for Environmental Information (NCEI) and Météo France. We aim at predicting 20 conditional quantiles, equally spaced between $\tau = 0.05$ and $\tau = 0.95$. Given that load consumption is strongly dependent on the time of the day, it is common practice (e.g., Gaillard, Goude, and Nedellec 2016) to fit a different model for each half-hour. To limit the computational burden, here we consider only the period between 11:30 and 12 a.m. We use the period 2005–2009 of the GEFCom2014 data for training, the last two years for testing. Similarly, we test each method on the last 24 and 12 months of, respectively, the UK and the French dataset.

Gaillard, Goude, and Nedellec (2016) proposed a quantile regression method which ranked 1st on both the load and the price forecasting track of GEFCom2014. This is a two-step procedure, which was partially motivated by the lack of reliable software for fitting additive quantile models. Very briefly, their method firstly fits a Gaussian additive model to model mean load and a second one to model the variance of the residuals

from the first fit. Then, for each quantile, they fit a linear quantile regression to model the load, using the effects estimated by the Gaussian fits as covariates. We compare their method to our proposal and to gradient boosting, using the set of covariates proposed by Gaillard, Goude, and Nedellec (2016): hourly temperatures (T_t); smoothed temperature (T_t^s), obtained using $T_t^s = \alpha T_{t-1}^s + (1 - \alpha)T_t$ with $\alpha = 0.95$; a cyclic variable indicating the position within the year (S_t); a factor variable indicating the day of the week (D_t); a sequential index representing time (t); the observed load at the same time of the previous day (L_{t-48}). Hence, the model for quantile τ is

$$q_\tau(L_t) = \psi_{D_t} + f_1^{15,6}(T_t) + f_2^{15,6}(T_t^s) + f_3^{20,8}(S_t) + f_4^{10,4}(L_{t-48}) + f_5^{4,4}(t),$$

where all smooth effects are based on cubic regression splines and, for instance, $f_1^{15,6}$ indicates that we used a basis of rank 15 and 6 degrees of freedom (the latter need to be chosen only under boosting). ψ_{D_t} is a fixed effect, whose value depends on D_t .

To let the learning rate and the ELF loss smoothness depend on the covariates, and to implement the method of Gaillard, Goude, and Nedellec (2016), we need also a variance model for the preliminary location-scale Gaussian GAM fit. We use

$$\log \text{var}(\text{Load}_t) = \tilde{\psi}_{D_t} + \tilde{f}_1^{10}(T_t^s) + \tilde{f}_2^{10}(S_t),$$

where the degrees of freedom do not need to be chosen. We consider two version of ELF-based quantile regression, one where $\sigma(x) = \sigma_0$ and another where σ depends on the covariates, using the location-scale model just described. In both cases, we calibrate σ_0 using $\tilde{V}$ -based calibration, because for this application it gives exactly the same results as bootstrapping, but

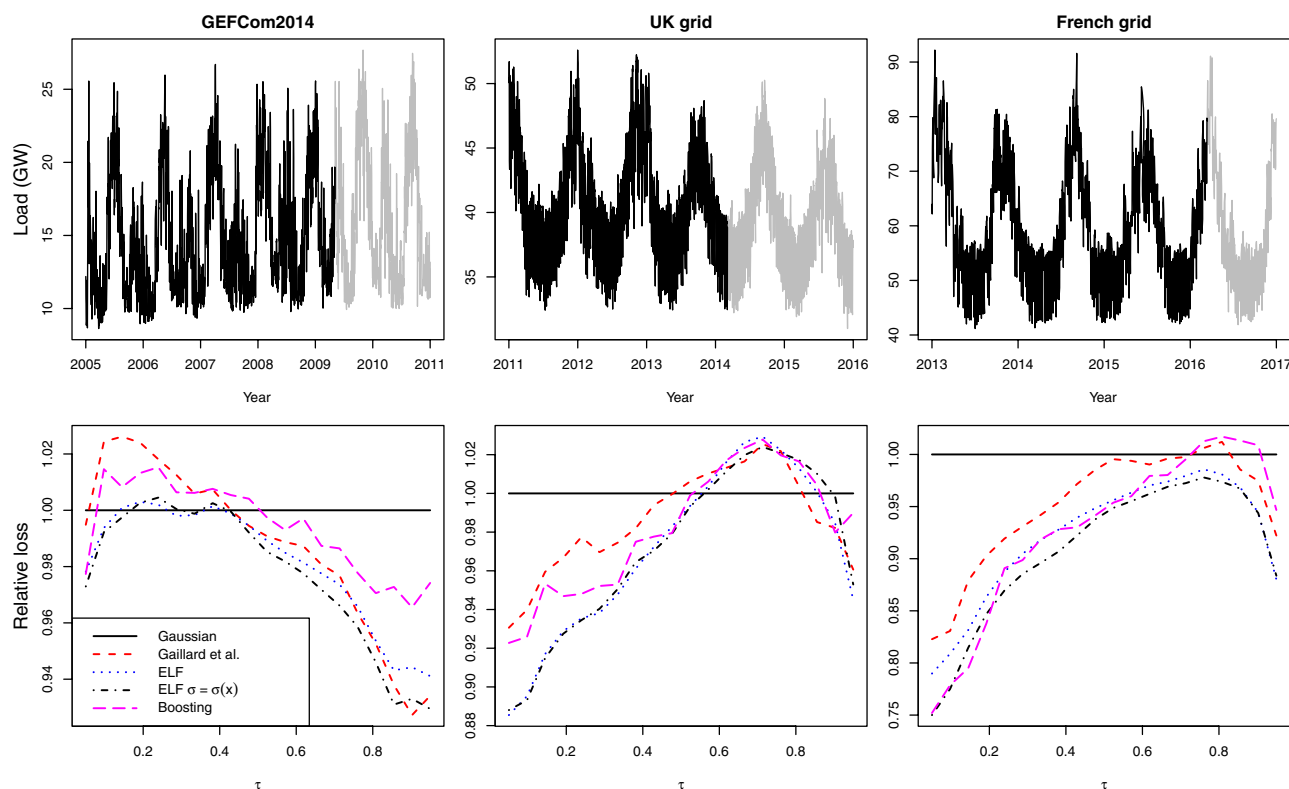


Figure 4. Top row: Daily electricity loads, observed between 11:30 and 12 a.m., from the GEFCom2014 challenge, the UK and the French grid, divided between training (black) and testing (gray) sets. Bottom row: Relative pinball losses for each τ and method.

at a much lower computational cost. For boosting, we use 100 bootstrap replicates to select the number of steps, while the step-size is equal to 0.1. Having tuned σ_0 and the number of boosting steps on the training sets, we forecast electricity load one week ahead, on rolling basis, using the test sets. See SM 4 for further details.

The bottom plots in Figure 4 show, for each τ and dataset, the pinball losses incurred on the testing sets, divided by the pinball loss of a Gaussian additive fit. Notice that both versions of ELF-based quantile regression do better than a Gaussian fit for most quantiles and that they are more accurate than the alternative quantile regression methods on each dataset. Remarkably, on the GEFCom2014 dataset, the proposed approach is superior to that of Gaillard, Goude, and Nedellec (2016), which was developed in the context of that challenge. Letting the learning rate σ depend on x leads to slightly improved performance on the GEFCom2014 and French data, but not on the UK data. On these data all quantile regression methods struggle to beat the Gaussian GAM around $\tau = 0.7$, which might be attributable to temperature anomaly that characterized the winter 2015/16 in the United Kingdom.

Regarding computing time, $\tilde{V}$ -based calibration takes around 2 sec on the GEFCom2014 training data. This includes the time needed to fit a Gaussian GAM, when $\sigma(x) = \sigma_0$. Fitting a location-scale Gaussian GAM takes around 20 sec, but this needs to be done only once, and the output is used for all quantiles. For gradient boosting, the number of steps which minimizes the cross-validated risk criterion varies widely across quantiles. In fact, for $\tau \approx 0.95$ more than 5400 steps are needed, while for $\tau \approx 0.4$ it is sufficient to use 550 steps. Using `mboost`, cross-validation takes around 800 sec in the first case and 80 sec

in the second. In practice, the optimal number of steps is not known in advance, hence it is necessary to cross-validate beyond the optimal point. We considered up to 1×10^4 steps for all quantiles, which translates to roughly 1450 sec. In the simulation setting considered here, where the smoothing parameters and regression coefficients are updated every week, our method has the further advantage that it is possible to initialize using the latest fit.

8. Conclusion

Motivated by the need for more flexible GAM models at EDF, this work provides a computationally stable and efficient framework for fitting additive quantile regression models. The learning rate and all smoothing parameters are selected automatically and calibrated uncertainty estimates are provided at no additional computational cost. This was achieved by coupling the general Bayesian belief updating framework of Bissiri, Holmes, and Walker (2016) with a statistically advantageous smooth generalization of the “pinball” loss and a novel efficient calibration procedure based on a Bayesian sandwich covariance estimator. Adopting the smooth ELF loss was key to computationally efficiency, because it allowed us to exploit the fast stable methods of Wood, Pya, and Säfken (2016), when selecting the smoothing parameter by marginal loss minimization. While working with a smooth loss is not novel in itself (see, e.g., Yue and Rue 2011; Oh, Lee, and Nychka 2012), the key contribution here is having selected the loss smoothness so as to minimize the asymptotic MSE of the estimated regression coefficients. Notice that the approach to ELF loss smoothness selection proposed

here applies to linear quantile regression models as well, while the new methods for learning rate calibration could be extended to loss-based Bayesian model fitting in general.

In practical terms the electricity load forecasting example demonstrates the practical utility of the proposed approach. Performance is better than that of gradient boosting, and at much lower computational cost (even more so when uncertainty estimates are required). Similarly, the methods are competitive with the ad hoc approach of Gaillard, Goude, and Nedellec (2016), on the very example that method was designed for. The methods proposed here fit each quantile model individually and do not impose any noncrossing constraint. Hence, future work might focus on extending the current framework to joint quantile fitting, with the aim of avoiding the quantile crossing problem, while maintaining interval calibration and computational efficiency.

Supplementary Materials

The online supplementary materials provide formal proofs of the theoretical results and computational details.

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